

O-RIS Assisted NOMA-VLC System: Integrated Positioning and Communications

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Abstract—This work introduces an optimized integrated positioning and communications (IPAC) framework for a non-orthogonal multiple access (NOMA) based visible light communication (VLC) system, enhanced by the assistance of an optical-reconfigurable intelligent surface (O-RIS). First, a closed-form expression is derived for the variance of the positioning estimation error in the considered set up using the Cramer-Rao Lower Bound. Next, we formulate a multi-objective optimization problem aimed at efficiently allocating the O-RIS elements among the communication and target users. In this context, our objective is two-fold: i) to minimize the derived positioning error metric for the target users; ii) to maximize the achievable sum rate for the communication users, while ensuring the quality-of-service requirements. To address this optimization problem, we employ a modified iterative greedy approach and demonstrate significant improvements both in terms of the achievable sum rate and the positioning error. Indicatively, at a meaningful for VLC transmit signal-to-noise ratio of 140 dB, the sum rate increases by over 9.5%, whilst the positioning error reduces by more than 72% compared to non O-RIS systems.

Index Terms—Integrated positioning and communication (IPAC), NOMA, optical-RIS, VLC systems, optimization, localization.

I. INTRODUCTION

THE EVER-INCREASING demand for high-speed, reliable and energy efficient wireless sensing and communication services has driven the exploration and integration of innovative solutions for next-generation communication systems [1]. Among these solutions, visible light communication (VLC) has emerged as a promising technology that utilizes light-emitting diodes (LEDs) for simultaneous data transmission and illumination. However, communication through visible light is susceptible to blockage, which is pronounced in indoor environments where the movement of people and objects often obstructs direct line-of-sight (LoS) links, compromising the reliability of both communication and positioning. Hence, optical reconfigurable intelligent surfaces (O-RIS) have been proposed to enhance the coverage performance and mitigate blockage issues by creating additional links [2]. By intelligently processing reflected signals, O-RIS can significantly improve signal quality, coverage, and capacity.

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To support efficient communication among multiple users in VLC systems, various multiple access techniques have been introduced, with non-orthogonal multiple access (NOMA) constituting a particularly promising approach. Power domain NOMA, in particular, multiplexes multiple users in the power domain allowing them to simultaneously share the same time, frequency, and code resources. This significantly increases spectral efficiency, improves connectivity, and makes NOMA a promising technology for congested indoor environments. To achieve that, power domain NOMA utilizes superposition coding at the transmitter and successive interference cancellation (SIC) at the receiver. Additionally, NOMA has demonstrated superior performance when compared to orthogonal multiple access (OMA) techniques in VLC systems [3], largely due to the disparity between the channel gains among users. To this effect, O-RIS can be also utilized for simulating the channel gain difference, resulting to enhanced system performance. For instance, a significant sum-rate improvement was achieved in [4], but not in the context of integrating positioning and communications.

While NOMA-supported VLC systems have unlocked a range of applications such as indoor positioning, intelligent lighting, and wireless data communication, certain challenges remain in achieving accurate positioning and optimizing system performance. The integration of positioning and communications (IPAC) in VLC systems has been explored in [5], [6], [7], [8]. These studies highlighted the potential benefits of IPAC systems, including enhanced system capacity, improved positioning accuracy, and reduced interference. For instance, an IPAC system for VLC with multiple users and access points was proposed in [6], employing a hybrid orthogonal frequency-division multiple access (OFDMA) and received signal strength (RSS) algorithm to reduce out-of-band interference and enhance positioning accuracy. However, such approaches typically require simultaneous capture of signals from at least two transmitters, which may not be feasible in all environments such as stairs and tunnels. To address this, a single-transmitter IPAC-VLC system utilizing on-off keying (OOK) modulation was introduced for a single user in [8], optimizing the positioning error variance using the Cramer-Rao lower bound (CRLB) and ensuring quality-of-service (QoS) requirements.

To the best of our knowledge, the present contribution is the first to exploit NOMA and O-RIS technologies for IPAC-VLC systems. To that end, we formulate a multi-objective optimization problem for efficient allocation of O-RIS elements among communication and target users. Its objectives include minimizing the positioning error metric for target users and maximizing the achievable sum rate for communication users while ensuring QoS requirements. The main technical contributions of this work are:

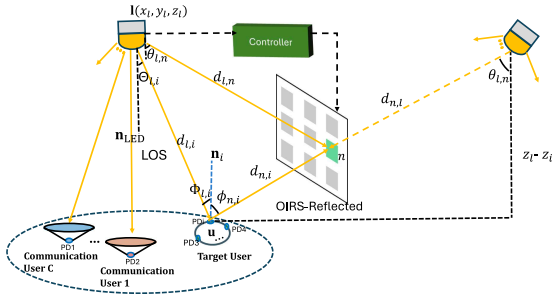


Fig. 1. O-RIS assisted integrated IPAC-VLC system.

- 1) A novel O-RIS-assisted IPAC framework for NOMA-VLC systems is proposed which is capable of supporting multiple users, while requiring only a single transmitter.
- 2) A closed-form expression for the positioning error variance in O-RIS-assisted IPAC-NOMA-VLC systems is derived using the CRLB, which serves as a performance metric.
- 3) To solve the formulated problem, we propose a modified iterative Greedy algorithm (IGA), with extensive simulations exhibiting significant improvements in both sum rate and positioning error accuracy compared to IPAC systems without O-RIS and sub-optimum O-RIS-assisted IPAC systems.

1) *Notations:* Let a , $\mathbf{a}$, $\mathbf{A}$, $\mathbf{a}^T$, and $\mathcal{A} \triangleq \{1, \dots, A\}$ denote the scalar value, vector, matrix, transpose of a vector, and set, respectively, whilst ∂ denotes gradient. Also, $\text{Tr}(\cdot)$ represents the trace function and $\text{Cov}(\cdot)$ is the covariance matrix. Finally, $\|\cdot\|_2$ denotes the 2-norm, and $\mathcal{G}$ denotes the Gaussian distribution.

II. SYSTEM MODEL

As shown in Fig. 1, an O-RIS-assisted NOMA-based IPAC framework for indoor VLC system is considered. The system consists of a single light-emitting diode (LED) source, $\mathcal{T}$ target users, and $\mathcal{C}$ communication users. For signal reception, each communication user utilizes M_c photodetectors (PDs) and each target user utilizes M_τ PDs. Therefore, the system has a total number of $I = \mathcal{C} \times M_c \times \mathcal{T} \times M_\tau$ PDs. Each PD has an area of A_P and a field-of-view angle of ψ . We assume that the positions of the communication users and the channel state information (CSI) are known, where CSI is obtained through adaptive statistical Bayesian minimum mean square error (MMSE) channel estimation techniques [9]. Also, an N -element O-RIS is utilized to create virtual LoS paths for both communication and target users. The positions of the LED, the n -th O-RIS element, the c -th communication user, and the τ -th target user are represented by $\mathbf{l} = [x_l, y_l, z_l]^T$, $\mathbf{r}_n = [x_n, y_n, z_n]^T$, $\mathbf{u}_c = [x_c, y_c, z_c]^T$, and $\mathbf{u}_\tau = [x_\tau, y_\tau, z_\tau]^T$, respectively, where $n \in \mathcal{N}$, $c \in \mathcal{C}$, $\tau \in \mathcal{T}$. Also, the position of the i -th PD, $i \in \mathcal{I}$, is denoted as $\mathbf{u}_i = [x_i, y_i, z_i]^T$. By leveraging the known positions of the M_τ PDs relative to the central reference point $\mathbf{u}$ and employing triangulation techniques [8] can determine the target user's location.

A. Channel Model

In indoor VLC, the LoS channel gain from the LED to the i -th PD is given by:

$$h_{l,i}^{\text{los}} = \frac{A_P(m+1)G_f G_c}{2\pi d_{l,i}^2} \cos^m(\Theta_{l,i}) \cos(\Phi_{l,i}), \quad (1)$$

where $m = -\ln 2 / \ln(\cos \theta_{1/2})$ is the Lambertian emission order with $\theta_{1/2}$ defined as the semi-angle at half power. Also, G_f and G_c are constants representing the gain from the optical filter and concentrator, respectively [10]. Thus, $d_{l,i}^2$ is the distance between the LED and the i -th PD, whilst $\Theta_{l,i}$ and $\Phi_{l,i}$ are the corresponding angles of irradiance and incidence, respectively. The cosine of these angles can be obtained from the geometry of Fig. 1 as $\cos(\Theta_{l,i}) = \cos(\Phi_{l,i}) = (z_l - z_i) / \|\mathbf{l} - \mathbf{u}_i\|_2$.

Similarly, under the assumption of the point source [11], [12], [13], the RIS-assisted channel gain from the LED to the i -th PD, which is reflected by the n -th O-RIS element, is given by

$$h_{l,n,i}^{\text{ris}} = \eta \frac{A_P(m+1)G_f G_c}{2\pi(d_{l,n} + d_{n,i})^2} \cos^m(\theta_{l,n}) \cos(\phi_{n,i}), \quad (2)$$

where η is the energy loss due to reflection through O-RIS. Thus, $d_{l,n}$ denotes the distance from the LED to the n -th O-RIS element and $d_{n,i}$ denotes the distance from n -th element of the O-RIS to the i -th PD. Furthermore, $\theta_{l,n}$, $\phi_{n,i}$, are the corresponding angles of irradiance and incidence, respectively. The cosine of these angles can also be obtained from the geometry of Fig. 1 as $\cos(\theta_{l,n}) = \cos(\phi_{n,i}) = (z_l - z_i) / \|\mathbf{l} - \mathbf{r}_n\|_2 + \|\mathbf{r}_n - \mathbf{u}_i\|_2$. Thus, the combined channel gain from the LED to the i -th PD is:

$$h_{l,i} = h_{l,i}^{\text{los}} + \sum_{n=1}^N h_{l,n,i}^{\text{ris}} \cdot \Omega_{n,i} = \beta \frac{(z_l - z_i)^{m+1}}{(\|\mathbf{l} - \mathbf{u}_i\|_2)^{m+3}} + \eta \beta \sum_{n=1}^N \frac{(z_l - z_i)^{m+1}}{(\|\mathbf{l} - \mathbf{r}_n\|_2 + \|\mathbf{r}_n - \mathbf{u}_i\|_2)^{m+3}} \cdot \Omega_{n,i}, \quad (3)$$

where $\beta = A_P(m+1)G_f G_c / 2\pi$. Additionally, the assignment indicator $\Omega_{n,i}$ takes a value of 1 when the O-RIS element n is assigned to the PD i , and 0 otherwise, i.e., $\Omega_{n,i} \in \{0, 1\}$. The O-RIS assignment matrix, whose elements are made up of $\Omega_{n,i}$, is represented by $\mathbf{S}$ as a matrix of size N by I .

B. Signal Model

In our system, NOMA is used to multiplex the signals of communication users in the power domain, while the target user utilizes this to estimate its own location. Specifically, the transmitter assigns different power levels to each communication user based on their individual channel conditions. Consequently, users with stronger channel gains are allocated lower power levels, while those with weaker channel gains are allocated higher power levels. On the receiver side, successive interference cancellation (SIC) is employed to facilitate the detection of the intended user's signal. Since the assignment of O-RIS elements can impact the overall channel gain, we adjust the ordering of the users based on the combined LoS and O-RIS reflected links, i.e., $h_{l,1} \geq h_{l,2} \geq \dots \geq h_{l,C}$ for all the channels. As a result, the assigned power allocation factors should satisfy $\alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_C$ and $\sum \alpha_c = 1$ for all $c \in \mathcal{C}$, where α_c represents the power allocation factor for the c -th user.

Let $s_c(t)$ represent the signal of the c -th communication user at time t . The amplitude of the signal is bounded by $A > 0$, i.e., $|s_c(t)| \leq A$, and $\mathbb{E}[s_c(t)] = 0$. Furthermore, we assume that the signal $s_c(t)$ is independent with variance of

$\mathbb{E}[s_c^2(t)] = \epsilon$. For on-off keying (OOK), the variance is $A^2/2$, for $t \in [0, T]$, where T is the period. Consider P_e being the LED electrical transmit power and $S(t) = \sum_{c=1}^C \sqrt{\alpha_c P_e} s_c(t) + I_d$ being the composite transmit signal also used for positioning and I_d represents the DC offset. The received signal at the i -th PD of a communication or target user is then given by $y_{l,i}(t) = h_{l,i}S(t) + n_i(t)$, where $n_i(t)$ is the additive white Gaussian noise (AWGN) at the i -th PD, which is assumed to have zero mean and σ^2 variance, i.e., $n_i \sim \mathcal{G}(0, \sigma^2)$. As the noise introduced by the photodetector is modeled as AWGN[8], the received signal $y_{l,i}(t)$ follows a Gaussian distribution with mean $h_{l,i}S(t)$ and variance σ^2 . This results in the following probability density function (PDF):

$$f(y_{l,i}(t); \mathbf{u}_j) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_{l,i}(t) - h_{l,i}S(t))^2}{2\sigma^2}}, \quad (4)$$

where $j = c$ for communication users and τ for target users.

III. PERFORMANCE ANALYSIS

A. Derivation of the Positioning Error Using CRLB

In this section, we derive a closed-form expression for the variance of the positioning estimation error of the target user (i.e., $j = \tau$) in the O-RIS-assisted NOMA-VLC system using the CRLB. Firstly, we obtain the log-likelihood function of the received signal over an observation period of T by [8]

$$\begin{aligned} \zeta(\mathbf{u}_\tau) &= \ln \left(\prod_{m=1}^{M_\tau} f(y_{l,m}(t); \mathbf{u}_\tau) \right) \\ &= \sum_{m=1}^{M_\tau} \int_0^T \left(-\frac{1}{2} \ln(2\pi\sigma^2) - \frac{(y_{l,m}(t) - h_{l,i}S(t))^2}{2\sigma^2} \right) dt \\ &= -\frac{M_\tau}{2} T \ln(2\pi\sigma^2) - \sum_{i=1}^{M_\tau} \int_0^T \frac{(y_{l,i}(t) - h_{l,i}S(t))^2}{2\sigma^2} dt. \end{aligned} \quad (5)$$

Let $\hat{\mathbf{u}}_\tau$ represent the estimated position of the target user, while $\mathbf{u}_\tau$ represents its actual position. The 3-D positioning error vector is then defined as $\mathbf{E}_\tau = \mathbf{u}_\tau - \hat{\mathbf{u}}_\tau = [e_x, e_y, e_z]^T$. We assume that $\mathbf{E}_\tau$ follows a Gaussian distribution[8] with zero mean and a covariance matrix $\text{Cov}(\hat{\mathbf{u}}_\tau)$. To measure the positioning accuracy, we utilize the CRLB and recall that the variance of any unbiased estimator of a parameter is at least as large as the inverse of the Fisher information matrix (FIM)[8], namely

$$\text{Tr}(\text{Cov}(\hat{\mathbf{u}}_\tau)) \geq \text{Tr}(\mathbf{F}_\tau^{-1}(\mathbf{u}_\tau)), \quad (6)$$

where $\mathbf{F}_\tau(\mathbf{u}_\tau)$ is the FIM associated with $\mathbf{u}_\tau$, which is calculated as the negative expectation of the second derivative of the log-likelihood function with respect to the parameters $\mathbf{u}_\tau = (x_\tau, y_\tau, z_\tau)$. This 3-D parameter vector tends to a 3×3 matrix or $a \times b$ matrix with $a \in \{1, 2, 3\}$ and $b \in \{1, 2, 3\}$. Also, $[\cdot]_{ab}$ represents the element located at the a -th row and b -th column of a matrix. As a result, the FIM can be given by

$$\begin{aligned} [\mathbf{F}_\tau(\mathbf{u}_\tau)]_{ab} &= -\mathbb{E} \left\{ \frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial u_{\tau a} \partial u_{\tau b}} \right\} \\ &= \begin{bmatrix} -\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial x_\tau \partial x_\tau} \right] & -\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial x_\tau \partial y_\tau} \right] & -\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial x_\tau \partial z_\tau} \right] \\ -\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial y_\tau \partial x_\tau} \right] & -\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial y_\tau \partial y_\tau} \right] & -\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial y_\tau \partial z_\tau} \right] \\ -\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial z_\tau \partial x_\tau} \right] & -\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial z_\tau \partial y_\tau} \right] & -\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial z_\tau \partial z_\tau} \right] \end{bmatrix}. \end{aligned} \quad (7)$$

Considering $\mathbb{E}[s_c(t)] = 0$ and $\mathbb{E}[s_c^2(t)] = \epsilon$, the only term in $\zeta(\mathbf{u}_\tau)$ that depends on x_τ , y_τ , and z_τ is the channel $h_{l,\tau}$, as shown in (5). To this effect, we compute the expectations of the partial derivatives terms of $h_{l,\tau}$ in (7) with respect to x_τ , y_τ , and z_τ , which are expressed as follows:

$$\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial x_\tau \partial x_\tau} \right] = -\frac{T}{\sigma^2} \sum_{i=1}^{M_\tau} \frac{\partial h_{l,i}}{\partial x_i} \frac{\partial h_{l,i}}{\partial x_i} (C^2 I_d^2 + P_e \epsilon), \quad (8)$$

$$\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial y_\tau \partial y_\tau} \right] = -\frac{T}{\sigma^2} \sum_{i=1}^{M_\tau} \frac{\partial h_{l,i}}{\partial y_i} \frac{\partial h_{l,i}}{\partial y_i} (C^2 I_d^2 + P_e \epsilon), \quad (9)$$

$$\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial z_\tau \partial z_\tau} \right] = -\frac{T}{\sigma^2} \sum_{i=1}^{M_\tau} \frac{\partial h_{l,i}}{\partial z_i} \frac{\partial h_{l,i}}{\partial z_i} (C^2 I_d^2 + P_e \epsilon), \quad (10)$$

$$\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial x_\tau \partial y_\tau} \right] = -\frac{T}{\sigma^2} \sum_{i=1}^{M_\tau} \frac{\partial h_{l,i}}{\partial x_i} \frac{\partial h_{l,i}}{\partial y_i} (C^2 I_d^2 + P_e \epsilon), \quad (11)$$

$$\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial y_\tau \partial z_\tau} \right] = -\frac{T}{\sigma^2} \sum_{i=1}^{M_\tau} \frac{\partial h_{l,i}}{\partial y_i} \frac{\partial h_{l,i}}{\partial z_i} (C^2 I_d^2 + P_e \epsilon), \quad (12)$$

$$\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial x_\tau \partial z_\tau} \right] = -\frac{T}{\sigma^2} \sum_{i=1}^{M_\tau} \frac{\partial h_{l,i}}{\partial x_i} \frac{\partial h_{l,i}}{\partial z_i} (C^2 I_d^2 + P_e \epsilon), \quad (13)$$

where the partial derivatives terms in (8)-(13) are derived as

$$\begin{aligned} \frac{\partial h_{l,i}}{\partial x_i} &= -\sum_{n=1}^N \beta \eta \Omega_{n,\tau} \frac{(m+3)(z_l - z_\tau)^{m+1}(x_n - x_\tau)}{(\|1 - \mathbf{r}_n\|_2 + \|\mathbf{r}_n - \mathbf{u}_i\|_2)^{m+4} (\|\mathbf{r}_n - \mathbf{u}_i\|_2)} \\ &\quad + \beta \frac{(m+3)(z_l - z_\tau)^{m+1}(x_l - x_\tau)}{(\|1 - \mathbf{u}_i\|_2)^{m+5}} \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{\partial h_{l,i}}{\partial y_i} &= -\sum_{n=1}^N \beta \eta \Omega_{n,\tau} \frac{(m+3)(z_l - z_\tau)^{m+1}(y_n - y_\tau)}{(\|1 - \mathbf{r}_n\|_2 + \|\mathbf{r}_n - \mathbf{u}_i\|_2)^{m+4} (\|\mathbf{r}_n - \mathbf{u}_i\|_2)} \\ &\quad + \beta \frac{(m+3)(z_l - z_\tau)^{m+1}(y_l - y_\tau)}{(\|1 - \mathbf{u}_i\|_2)^{m+5}} \end{aligned} \quad (15)$$

$$\begin{aligned} \frac{\partial h_{l,i}}{\partial z_i} &= \sum_{n=1}^N \beta \eta \Omega_{n,\tau} \left(\frac{-(m+1)(z_l - z_\tau)^m}{(\|1 - \mathbf{r}_n\|_2 + \|\mathbf{r}_n - \mathbf{u}_i\|_2)^{m+3}} \right. \\ &\quad \left. + \frac{(m+3)(z_l - z_\tau)^{m+1}(z_n - z_\tau)}{(\|1 - \mathbf{r}_n\|_2 + \|\mathbf{r}_n - \mathbf{u}_i\|_2)^{m+4} (\|\mathbf{r}_n - \mathbf{u}_i\|_2)} \right) + \beta \\ &\quad \times \left(\frac{-(m+1)(z_l - z_\tau)^m}{(\|1 - \mathbf{u}_i\|_2)^{m+3}} + \frac{(m+3)(z_l - z_\tau)^{m+2}}{(\|1 - \mathbf{u}_i\|_2)^{m+5}} \right). \end{aligned} \quad (16)$$

By substituting from (14)-(16) in (8)-(13) and then (7) yields

$$[\mathbf{F}_\tau(\mathbf{u}_\tau)]_{ab} = -\mathbb{E} \left[\frac{\partial^2 \zeta(\mathbf{u}_\tau)}{\partial u_{\tau a} \partial u_{\tau b}} \right] = -\frac{(C^2 I_d^2 + P_e \epsilon) T}{\sigma^2} \mathbf{V}(\mathbf{u}_\tau), \quad (17)$$

where $\mathbf{V}(\mathbf{u}_\tau)$ represents the matrix containing the dependent terms on $\mathbf{u}_\tau$. Thus, (6) can be expressed as

$$\text{Tr}(\text{Cov}(\hat{\mathbf{u}}_\tau)) \geq \frac{\sigma^2}{(C^2 I_d^2 + P_e \epsilon) T} \text{Tr}(\mathbf{V}^{-1}(\mathbf{u}_\tau)). \quad (18)$$

B. Sum Rate

Sum rate, R_t , is an effective metric for assessing users' performance. Based on this and since each user employs a single PD (i.e., $M_c = 1$ and $h_{l,i,c} \rightarrow h_c$), it follows that [4], [12]:

$$R_t = \sum_{c=1}^C R_c = \sum_{c=1}^C B \log_2 \left(1 + \frac{e}{2\pi} \gamma_c \right), \quad (19)$$

where B is the bandwidth and γ_c is the signal-to-interference-plus-noise ratio (SINR) of the c -th user. Assuming perfect SIC and that the first user's signal is demodulated at the end

so it will not be interfered with the remaining users' signals yields [4]:

$$\gamma_c = \begin{cases} \gamma_0 |h_1|^2 \alpha_1, & \text{for } c = 1 \\ \frac{|h_c|^2 \alpha_c}{\frac{1}{\gamma_0} + |h_c|^2 \sum_{k=1}^{c-1} \alpha_k}, & \text{for } 2 \leq c \leq C. \end{cases} \quad (20)$$

where $\gamma_0 = P_e/\sigma^2 = P_e/(N_0 B)$ is the transmit SNR with N_0 denoting the corresponding noise spectral density.

C. Problem Formulation

In this section, we formulate an optimization problem to obtain the O-RIS assignment matrix ($\mathbf{S}$) that jointly minimizes the variance of the positioning estimation error of the target users and maximizes the sum rate of the communication users. The problem is formulated as:

$$\begin{aligned} \max_{\mathbf{S}} \quad & \sum_{c=1}^C \lambda R_c - (1 - \lambda) \text{Tr}(\mathbf{V}^{-1}(\mathbf{u}_\tau)) \\ \text{subject to:} \quad & R_c \geq R_{\text{th}}, \forall c \in \mathcal{C} \\ & \mathbf{E}_\tau \leq \mathbf{E}_{\tau_{\text{th}}}, \forall \tau \in \mathcal{T} \\ & \sum_{i=1}^I \Omega_{n,i} = 1, \forall n \in \mathcal{N} \\ & \Omega_{n,i} \in \{0, 1\}, \forall n \in \mathcal{N}, i \in \mathcal{I}, \end{aligned} \quad (21)$$

where λ is the regularization parameter and R_{th} is the threshold data rate. The first and second constraints in (21) guarantee the QoS requirement for each user. The third constraint is set to ensure that each O-RIS element is assigned to only one PD.

D. Problem Solving

The optimization problem in (21) is a non-convex multi-objective one. To solve it, a modified iterated greedy algorithm (IGA) is employed as a robust heuristic approach [14]. IGA is known for its effectiveness in solving complex optimization problems by iteratively improving the solutions based on a greedy strategy [14]. As given in Algorithm 1, it explores the search space and iteratively constructs candidate solutions by selecting and modifying components to optimize the multiple objectives simultaneously. This approach guarantees a faster convergence to a satisfactory solution with relatively low computational complexity.

IV. SIMULATION RESULTS

This section presents the simulation results to evaluate the effectiveness of the proposed O-RIS-assisted NOMA-IPAC indoor VLC system. Without loss of generality, the room dimensions are assumed to be $(8 \times 8 \times 3 \text{ m}^3)$, with a single LED (transmitter) fixed at coordinates $(x_l, y_l, z_l) = (4, 4, 3)$. Two communication users are considered ($C = 2$), located at coordinates $(x_c, y_c, z_c) = (4, 3, 0.5)$ and $(5, 3, 0.5)$. A single target user ($\mathcal{T} = 1$) is located at coordinates $(x_\tau, y_\tau, z_\tau) = (2.5, 2.5, 1)$. Each communication user is equipped with $M_c = 1$ PD, while each target user utilizes $M_\tau = 3$ PDs, each with $A_P = 1 \text{ cm}^2$ and $\psi = 80^\circ$. The O-RIS unit is placed at coordinates $(1, 0, 1.5)$ and consists of N elements, each with a size of $(0.10 \times 0.10 \text{ m}^2)$ and $\eta = 0.95$. The parameters G_f , G_c , and m are all set to unity. The power allocation factors are chosen as $\alpha_1 = 0.25$ and $\alpha_2 = 0.75$. Finally, we consider

Algorithm 1 IGA for O-RIS Assignment

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1: Input:  $C, \mathcal{T}, M_c, M_\tau, \lambda, R_{\text{th}}, \mathbf{E}_{\tau_{\text{th}}}, \varepsilon$ , and  $t_{\text{max}}$ 
2: Initialize: Assignment matrix  $\mathbf{S}$  with zeros and iteration counter  $t \leftarrow 0$ 
3: repeat
4:    $t \leftarrow t + 1$ 
5:   for  $n = 1$  to  $N$  do
6:     for  $i = 1$  to  $I$  do
7:        $\Omega_{n,i} \leftarrow i$ 
8:       Compute  $R_c, \mathbf{V}^{-1}(\mathbf{u}_\tau)$ ,  $\mathbf{E}_\tau$  based on  $\mathbf{S}$ 
9:       Objective  $\leftarrow \sum_{c=1}^C \lambda R_c - (1 - \lambda) \text{Tr}(\mathbf{V}^{-1}(\mathbf{u}_\tau))$ 
10:    end for
11:  end for
12:   $\hat{S} \leftarrow \arg \max(\text{Objective})$ 
13:  if  $R_c \geq R_{\text{th}}$  and  $\mathbf{E}_\tau \leq \mathbf{E}_{\tau_{\text{th}}}, \forall c \in \mathcal{C}, \forall \tau \in \mathcal{T}$  then
14:    break
15:  else
16:    Re-assign 5% ORIS elements to the unmet user
17:  end if
18:  Check termination condition ( $t_{\text{max}}$  or convergence)
19: until termination condition is met
20: Output: Final assignment matrix  $\mathbf{S}$ 

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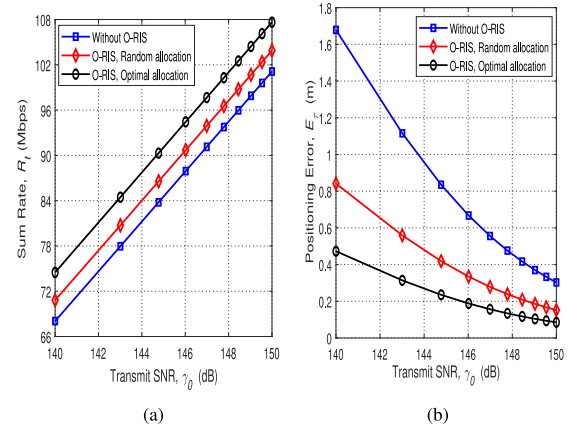


Fig. 2. Performance comparison between the proposed optimized framework and random allocation method: (a) Sum Rate (b) Positioning Error.

$N_0 = 10^{-21} \text{ A}^2/\text{Hz}$, $B = 20 \text{ MHz}$, $N = 150$, $I_d = 700 \text{ mA}$, $R_{\text{th}} = 10 \text{ Mbps}$, $\mathbf{E}_{\tau_{\text{th}}} = 0.85 \text{ m}$, and $\lambda = 0.5$.

Fig. 2 compares the sum rate (Fig. 2.a) and positioning error (Fig. 2.b) between the proposed IGA and two other schemes: the random allocation method and the system without O-RIS. It is observed that integrating O-RIS elements leads to a significant performance improvement for both metrics. This improvement is attributed to the additional received gain from the allocated O-RIS elements, as indicated in (3). Furthermore, the proposed optimized framework with IGA scheme outperforms the random allocation method with a 5% gain in sum rate and a substantial 43% reduction in positioning error at $\gamma_0 = 140 \text{ dB}$, which unlike in RF it is a typical value in VLC. This superiority is attributed to the optimized framework of the proposed IGA, which takes into account both the sum rate and positioning error. As a result, extra O-RIS elements are assigned to either the communication or target users based on their achievable performances, ensuring that both metrics

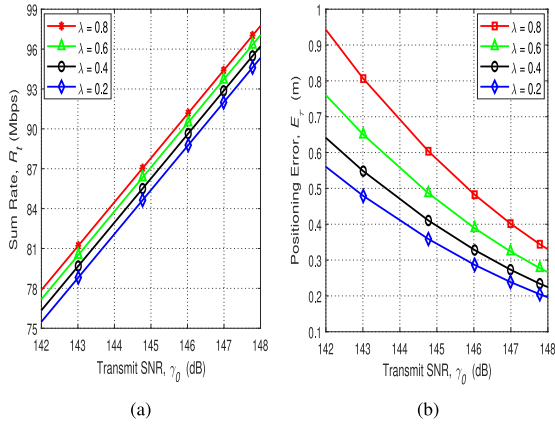


Fig. 3. The effect of the regularization parameter on the system performance: (a) Sum Rate, (b) Positioning Error.

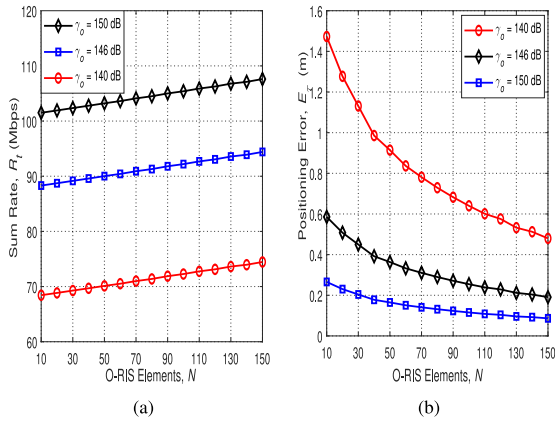


Fig. 4. The effect of the number of O-RIS elements on the system performance: (a) Sum Rate, (b) Positioning Error.

satisfy the threshold values. For instance, when considering $\gamma_0 = 143$ dB and the system without O-RIS deployment, the achievable sum rate and positioning error are $R_t = 77.9$ Mbps and $E\tau = 1.11$ m, respectively. These values improve to $R_t = 80.73$ Mbps and $E\tau = 0.558$ m for the random allocation method and are further enhanced to $R_t = 84.45$ Mbps and $E\tau = 0.324$ m for the proposed framework.

Fig. 3 examines the proposed framework and explores the impact of varying λ on the sum rate (R_t) and positioning error ($E\tau$). It is evident that increasing λ yields a positive effect on R_t of the communication users, while conversely impacting the $E\tau$ of the target users. This behavior stems from the formulation of our optimization problem in (21), where λ weight is assigned to R_t and $(1 - \lambda)$ weight is assigned to $E\tau$. For instance, consider a high λ value of 0.8 and $\gamma_0 = 140$ dB. In this case, the achievable sum rate and positioning error values are $R_t = 71.2$ Mbps and $E\tau = 1.21$ m, respectively. On the contrary, if we consider a low λ value of 0.2 with the same $\gamma_0 = 140$ dB, R_t is degraded to 68.9 Mbps, while $E\tau$ improves to 0.72.

Fig. 4 investigates the effect of the number of O-RIS elements (N) on the sum rate (R_t) and positioning error ($E\tau$) illustrating the trade-off between them. The figure depicts the trade-off between these two performance metrics as N varies. Thus, as N increases, the algorithm dynamically allocates these elements to optimize the R_t and $E\tau$, ensuring that both metrics

meet the desired thresholds R_{th} and $E\tau_{th}$, respectively. For instance, let's consider a system with $\gamma_0 = 140$ dB. When there are 50 elements, the algorithm achieves a R_t of 70.12 Mbps and a $E\tau$ of 0.16 m. As N increases to 100, $E\tau$ further improves to 0.11 m, while the R_t becomes 72.2 Mbps.

V. CONCLUSION

This work introduced a novel O-RIS-assisted NOMA-IPAC indoor VLC framework. where the corresponding positioning error of the target user were analytically quantified using CRLB. By considering both the sum rate and the developed positioning error metrics, a multi-objective optimization problem was then formulated and efficiently solved using an IGA. The performance of the proposed framework has been compared with two benchmark scenarios: the system without O-RIS deployment and the O-RIS-assisted system with a random allocation method. The achieved results have demonstrated the superiority of the proposed framework, showcasing its effectiveness in enhancing both sum rate and positioning error. Furthermore, the impact of key parameters, including the regularization parameter, number of O-RIS elements, and transmit SNR, has been thoroughly investigated. The adaptability of the proposed framework has been highlighted, as it dynamically allocates O-RIS elements based on the specific requirements of the system. This enables the system to balance the trade-off between sum rate and positioning error, ensuring that both metrics satisfy the desired thresholds.

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